Seamus Taylor

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PROFILE

Third-year Computer Science student at Monash specialising in Data Science, with two years building and leading applied AI systems. Shipped an LLM extraction pipeline with its own evaluation harness, architecting a clinical decision agent whose outputs stay auditable, and building the vision pipeline for an autonomously controlled prosthetic arm. Team Lead of Monash AIM for 2027. Looking for an AI engineering placement where the work has to survive contact with real users.

EDUCATION

Bachelor of Advanced Computer Science (Honours) — Monash University *2024 – 2027 (expected)*

Specialising in Data Science; minoring in Mathematics and Games Development.

EXPERIENCE

Team Lead (2027), Operations Lead (2026) — Monash AIM — Analysis of Images in Medicine *2025 – Current*

Monash's medical AI research group. We enter MICCAI, the leading international medical imaging conference, whose challenges are competitive benchmarks scored on held-out test sets against labs worldwide.

CHIMERA-Agent — clinical decision agent

- Building a Python risk-stratification pipeline over MRI and biopsy data, architected as a hybrid system: a deterministic guideline decision core handles the clinical logic, with a grounded LLM constrained to what it can actually support from the record.
- Risk-group derivation against published clinical guidelines returns candidate categories with traces rather than hard labels, so every output can be audited back to the evidence that produced it — the difference between a system you can deploy and one you can only demo.
- Automated the technical and advisory reporting the system produces; ran the project kickoff, setting the scoring design and build plan.

CohortX — LLM extraction pipeline (Team Lead, four people, three tasks)

- Led delivery of a pipeline extracting structured cohort selection criteria from biomedical free-text, and personally delivered two of the three tasks end to end: parsing, prompting, eligibility prediction, evaluation and submission.
- Selected open-weight models — LLaMA 3.1 8B as primary, Mistral Nemo and BioBERT as fallbacks — evaluated against accuracy, inference cost and the compute actually available.
- Built a local evaluation harness reimplementing the challenge's composite scoring metric, so the team could measure a change without burning a submission, and diagnosed two silent faults: a parsed-versus-expected file mismatch, and scorer weights that could not be verified.
- Restructured ownership and handover so the team kept delivering to deadline through a period of absence. The third-task result earned a contribution to a peer-reviewed MICCAI paper and presentation, in preparation.

Programme operations

- Coordinated five parallel challenge teams — allocation, delegation, deadline tracking, liaison with academic advisors — and fitted training workloads to a zero-budget constraint across Monash M3/MASSIVE HPC and free Kaggle and Colab tiers. One of the five won its challenge outright.

AI Lead, Computer Vision — NeuroGrip — Monash DeepNeuron *2026 – Current*

- Own the computer vision pipeline on a three-person build (AI, mechanical, electrical) working toward a prosthetic arm controlled autonomously by AI; my module detects the target object and drives finger movement from what the camera sees.
- Built monocular depth estimation on Depth Anything with an optional stereo path and the calibration for both; the pipeline tracks a taught fingertip and the target, estimates the gap in millimetres, and emits a movement direction to the electrical controller over a shared-memory and Ethernet bus.
- Deployed to an NVIDIA Jetson NEON smart camera for on-device inference; wrote synthetic scene generation so the pipeline could be tested and regression-checked before any hardware existed.
- Enforced a safety interlock as a design rule: no actuator command is sent until calibration and validation pass, so a miscalibrated model cannot drive a physical hand.

Research Collaborator — NeuroBOLT reimplementation — Independent collaboration *2026 – Current*

- Reimplementing NeuroBOLT (NeurIPS 2024) from scratch with a researcher — a transformer translating EEG into fMRI brain-activity signals via multi-scale spectral feature fusion.

SKILLS

Languages: Python, C++, SQL, R, basic Java

AI engineering: Open-weight LLMs (LLaMA, Mistral, BioBERT), structured output, grounded generation, hybrid deterministic/LLM architectures, evaluation harness design

ML & systems: PyTorch, transformers, CNNs, depth estimation, edge inference (Jetson), data pipelines, M3/MASSIVE HPC, Linux, Bash, Git

Leading: Team and operations leadership across five concurrent projects

Also: Web Developer, Centaur Association (Feb 2025 – current) — sole developer, full site rebuild, non-technical stakeholders